

RUTLAND SOUTHERN VERMONT, VT, USA

WMO: 725165

Lat: 43.533N

Lon: 72.950W

Elev: 239

StdP: 98.49

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.0	-29.0	0.3	-16.3	-26.0	0.4	-14.9	11.9	1.3	10.8	0.4	2.2	130	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.5	21.4	27.9	20.4	27.1	19.7	22.9	27.7	21.9	26.4	20.9	25.1	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.4	25.7	20.2	15.4	24.8	19.1	14.3	23.8	68.7	27.5	65.0	26.5	61.7	25.7	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.3	7.4	-25.5	32.2	3.4	1.4	-28.0	33.2	-30.0	34.0	-31.9	34.8	-34.4	35.9		
			-25.8	24.5	3.2	1.1	-28.1	25.3	-30.0	25.9	-31.8	26.6	-34.1	27.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-6.3	-4.7	-0.2	6.6	13.2	17.8	20.4	19.3	15.5	9.2	3.2	-2.5
	DBStd	10.47	7.17	6.11	5.77	4.94	4.33	3.57	2.91	2.99	4.04	4.53	5.21	5.68
	HDD10.0	2052	506	411	320	125	17	0	0	0	3	71	212	388
	HDD18.3	4080	763	644	574	352	169	52	13	25	103	284	455	646
	CDD10.0	1209	1	0	3	24	114	234	322	288	170	46	7	1
	CDD18.3	197	0	0	0	1	9	36	76	54	19	1	0	0
	CDH23.3	1554	0	0	3	25	109	314	594	365	138	6	0	0
	CDH26.7	332	0	0	1	4	21	72	141	67	27	0	0	0
Wind	WSAvg	3.2	3.3	3.5	3.5	3.8	3.3	2.9	2.6	2.5	2.8	3.1	3.3	3.2
Precipitation	PrecAvg	1148	81	67	87	89	94	113	120	104	97	113	93	93
	PrecMax	1363	157	160	159	175	160	262	223	188	216	266	140	161
	PrecMin	844	28	28	38	21	30	34	49	34	48	12	59	40
	PrecStd	154	32	29	30	35	37	50	48	39	37	67	26	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	13.6	21.0	27.0	28.9	31.3	32.0	31.0	29.9	24.0	19.1	14.0
		MCWB	11.5	8.7	13.3	15.9	19.1	22.3	23.6	22.2	21.3	16.9	14.1	10.0
	2%	DB	9.1	9.3	14.7	22.3	26.6	28.7	29.6	28.0	27.1	21.8	16.9	10.9
		MCWB	6.7	5.9	9.1	12.9	17.0	21.0	21.9	20.7	19.7	16.4	12.2	8.1
	5%	DB	6.1	6.4	11.2	18.7	23.8	27.1	27.9	27.2	24.7	19.0	13.8	7.7
		MCWB	3.4	3.1	6.5	11.5	15.7	19.3	21.0	20.2	18.9	14.5	10.2	5.6
	10%	DB	2.9	3.7	7.8	15.9	21.7	25.0	27.1	25.9	22.5	17.1	11.3	5.1
		MCWB	0.9	1.1	4.2	9.6	14.6	18.1	20.2	19.3	17.6	12.8	8.1	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	10.2	13.7	17.3	20.5	23.4	24.6	23.4	22.6	19.4	15.5	10.8
		MCDB	14.2	12.6	19.3	23.9	27.1	29.3	30.0	28.0	27.0	22.1	18.1	13.3
	2%	WB	7.3	6.2	10.1	14.5	18.9	22.0	23.3	22.2	21.0	17.0	12.8	8.2
		MCDB	8.8	8.9	13.7	20.7	24.2	26.9	28.1	26.7	25.1	20.1	15.6	9.6
	5%	WB	3.6	3.4	6.9	12.1	17.2	20.6	22.1	21.2	19.7	15.1	10.8	5.7
		MCDB	5.6	5.9	10.2	17.8	21.9	25.1	26.7	25.4	23.5	18.6	13.4	7.9
	10%	WB	1.2	1.4	4.5	9.8	15.6	19.3	21.1	20.2	18.5	13.1	8.1	3.1
		MCDB	3.0	3.7	7.9	14.9	20.3	23.4	25.3	24.2	21.8	16.4	10.6	4.8
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.3	9.4	10.7	11.2	10.7	10.6	10.6	10.6	9.5	8.2	7.6
		MCDBR	10.9	11.4	14.1	16.4	15.0	13.3	11.9	11.6	12.6	13.2	11.7	10.0
		MCWBR	9.6	9.0	9.4	9.5	8.0	7.0	6.1	6.1	6.9	8.3	8.9	8.8
	5% WB	MCDBR	10.4	10.7	12.6	14.3	12.6	11.2	10.5	10.1	10.7	11.4	10.3	9.8
	MCWBR	9.5	9.0	9.4	9.3	7.5	6.4	5.8	5.7	6.4	8.0	9.0	9.1	
Clear-Sky Solar Irradiance	taub	0.290	0.297	0.322	0.346	0.379	0.422	0.438	0.414	0.365	0.337	0.306	0.283	
	taud	2.361	2.319	2.384	2.414	2.363	2.255	2.216	2.271	2.398	2.487	2.519	2.463	
	Ebn at Noon	840	898	919	919	893	852	832	841	860	846	825	821	
	Edh at Noon	79	98	105	111	120	134	138	126	103	83	68	66	
All-Sky Solar Radiation	RadAvg	1.55	2.42	3.57	4.46	5.04	5.44	5.62	5.08	4.11	2.52	1.69	1.20	
	RadStd	0.15	0.21	0.26	0.52	0.58	0.47	0.41	0.27	0.31	0.27	0.18	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	N/A

Nomenclature: See separate page